PANTONE REPORT : THE PAGES OF MICHEL DAVID

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PERSONAL REFLECTIONS ON THE THEORETICAL OPERATION OF THE PANTONE REACTOR.

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What actually happens in the PANTONE reactor is quite controversial.

For my part, I think, that the whole matter rests on the fact that the fuel vapor which one forces to pass in the small interval between the central rod and the tube, is electrified by friction on this rod and the wall of this tube. It is essential that this electrification be obtained, otherwise the reactor is only a heat exchanger which at most allows the fuel to volatilize, which has already been experimented with in the past and only allows a slight improvement in the carburation. by better mixing with air.

On the other hand, if this fuel vapor electrifies, it decomposes into a combustible gas with very different characteristics from simple fuel vapor. This gas is therefore obtained when the conditions are met in the construction of the reactor and especially of its feed. The composition of this gas varies depending on the fuels introduced into the reactor, and are always based on molecules of hydrogen, oxygen, carbon, etc.

So Mr. PANTONE designed his reactor simply as a two-way heat exchanger. The exhaust gases circulate in a direction to heat the reactor, and the fuel, vaporized or volatilized, circulates in the opposite direction in the small space which is intended for it. The zone, in which the fuel vapor

circulates, is quite long, about 30 cm. In the first few cm, the atomized fuel transforms on contact with the hot walls into dry vapor, then the micro droplets of this dry vapor, continuing their advance in the small space between rod and tube, electrify by friction, which occurs easily if they travel fast enough and the tube and rod are not wet with fuel.

It should be noted that all these micro droplets and these molecules, barely charged with static electricity, will discharge immediately when approaching the walls or between them. They attract, they repel, they charge, they unload etc.

What is important to understand is that with each charge and discharge, a small spark occurs, this spark promotes the fragmentation of the droplets up to the molecule to see up to the atom.

All these sparks produce the famous rise in temperature which is observed towards the outlet of the reactor, a temperature which is higher than that of the exhaust gases.

This is a simplified approach to the phenomenon which is surely more complex and slightly variable depending on the products used. It is therefore necessary to consider that precombustion may occur in certain cases in the reactor and give rise to products different from those caused by electrostatic fragmentation.

We now understand the importance of obtaining and maximizing the electrostatic phenomenon of which I have just spoken.

Therefore, if the fuel is not sprayed finely enough before it is introduced into the reactor, or if the latter is not hot enough, the interior of the latter will be "wet" and there will be little or no appearance of electrostatic charge. Thus the reactor will not produce the desired result and the engine will operate in the traditional way because it will not be supplied by this very particular gas produced normally by the reactor.

Note that if you feed your reactor by the "bubbler" system rather than a carburetor or an injector, the electrification will be done very quickly at start-up as soon as the reactor begins to heat up, which prevents too much condensation, therefore the wetting of it initially, because the fuel is much more divided by the bubbling and has less chance of condensing. It is for this reason that the experimental systems carried out with "bubblers" work particularly well with volatile fuels, even if they are of rudimentary construction.

You must understand now, that it is imperative to respect three conditions, so that the invention of Mr PANTONE, produces all its effect:

- **1 - Prepare the fuel by volatilizing it or by pulverizing it as finely as possible before its introduction into the reactor.**
- **2 - Heat the reactor, so that the volatilized or pulverized fuel does not condense on contact with the rod and the internal wall of the tube which contains it. In the event of condensation, this, by "wetting" these elements, would prevent any electrification.**
- **3 - Ensure that the electrification (static electricity charging the atomized fuel and the internal walls of the reactor) is as intense as possible. To obtain this result, it is imperative that the speed of circulation in the reactor is as high as possible at all engine speeds.**

Obviously this is the first and last condition, which is the most difficult to achieve when the engine is running at low speed. I have spoken at length about this issue in my other pages.

WHAT IS THE USEFULNESS OF THE ELECTRIFICATION PHENOMENON IN THE OPERATION OF THE PANTONE REACTOR?

Consider what happens to a single micro droplet of hydrocarbon or water, which circulates at high speed, in the narrow space between the central rod and the inner wall of the reactor tube.

Given the length of the reactor, there will inevitably come a time when this droplet will "rub" against the wall or the rod. If the conditions mentioned above are met, this droplet will acquire a positive or negative electric charge contrary to that of the tube and the rod. Hardly acquired, the charge of this micro droplet will change polarity, either by a new contact with the wall or the rod which it attracts, since of opposite polarity, or in contact with a neighboring microdroplet which would also temporarily be of polarity. opposite.

As I said before, all these charges and discharges of static electricity producing micro sparks, raise the temperature of the reactor beyond that which can be produced by the exhaust gases.

The sparks also promote the fragmentation of the microdroplets until complete decomposition into molecules or even atoms, all this in a precombustion environment with all that may result from it.

I think that chemists specializing in cracking questions, could give you more explanations, supported by the corresponding calculations.

Now we know enough to work on the PANTONE reactor and to determine its dimensions, its shape, according to the intended use. Personally, I have made reactors in glass, copper, Teflon, according to the system of the tube and the central rod, but also others, completely flat.

It should also be borne in mind that the electrostatic charge develops more or less easily depending on the nature of the droplets and the nature of the internal walls of the reactor.

Thus, if it is hydrocarbon droplets, the electric charge will easily occur because we are dealing with a very insulating product, even at low speed and even in contact with a conductive material such as steel. When it comes to water vapor, the electric charge of molecules will have difficulty arising by friction in contact with steel, because we are dealing with two more or less conductive products.

I noticed that the operation of an engine equipped with a PANTONE reactor, was improving over time. I think it is due to the fact that the internal walls of the tube and the surface of the rod, covered each other little by little, with a very thin layer of an insulating product which favors the production of the electric charge.

Now you know the three main criteria for the proper functioning of a PANTONE reactor, in particular the need for a high gas velocity in it, which makes it almost essential, in some cases, to install a shutter device. momentary from the air intake located at the reactor outlet and directly on the intake (see previous pages).

With this device, after preheating, all gasoline engines run perfectly on fuel oil, that is to say they become gas engines.

To solve the problem of the speed of the gases, it would be interesting to design a reactor of variable size, which would be very small at idle, and would grow with the increase in engine speed.

In fact, it would be interesting to also remove the air intake, which poses so many problems for us, and to pass the air through the variable reactor, with the fuel.

I am currently experimenting with this type of reactor, and will keep you posted if the results are good.

DAVID M.

As early as 1840, we knew how to makestatic electricity with steam as evidenced by these extracts from

"Small Treatise on Physics"

by MJ Jamin

Gauthier-Villars Paris - 1882

William Georges Armstrong, English inventor (1810-1871) is known for his construction of machines and cannons. Its cannon system was applied to the entire English army in 1858 which made it famous and rich. Besides his cannons and his hydroelectric machine, he invented various hydraulic pressure machines.

Machine Armstrong. — Un mécanicien anglais, occupé à réparer des fuites dans la chaudière d'une machine à vapeur, vit des étincelles et éprouva des secousses quand il tenait une de ses mains dans le jet de vapeur et qu'il touchait avec l'autre la soupape de sûreté. M. Armstrong, qui étudia ce phénomène, reconnut que la séparation des deux fluides se fait, non dans l'intérieur de l'appareil pendant l'ébullition, mais au point où la vapeur sort en frottant contre les parois du robinet. M. Faraday prouva ensuite que la vapeur ne donne point d'électricité quand elle est surchauffée et sèche, tandis qu'elle en produit abondamment quand on la fait passer, avant sa sortie, dans une boîte contenant de l'étoupe mouillée, où elle se charge de gouttelettes liquides. Ce sont ces gouttes qui s'électrisent en frottant contre les parois du bec.

Quand l'eau de la chaudière est distillée, le bec devient négatif et le jet de vapeur positif. L'eau ordinaire ou saline ne produit rien. Avec des essences ou des corps gras, la vapeur est négative et la chaudière positive. La production de l'électricité dépend aussi de la matière des becs; elle est nulle s'ils sont en ivoire, et abondante quand ils sont métalliques; le bois de buis donne les résultats les plus énergiques.

La machine Armstrong (*fig.* 221) est fondée sur ces propriétés. Elle se compose d'une chaudière remplie d'eau distillée qu'on porte à 6 atmosphères. La vapeur s'échappe à travers des becs C après avoir été refroidie dans une boîte pleine d'eau D, qui est représentée à part dans la *fig.* 222;

en sortant elle contourne une lame verticale de buis sur laquelle elle frotte

Fig. 221.

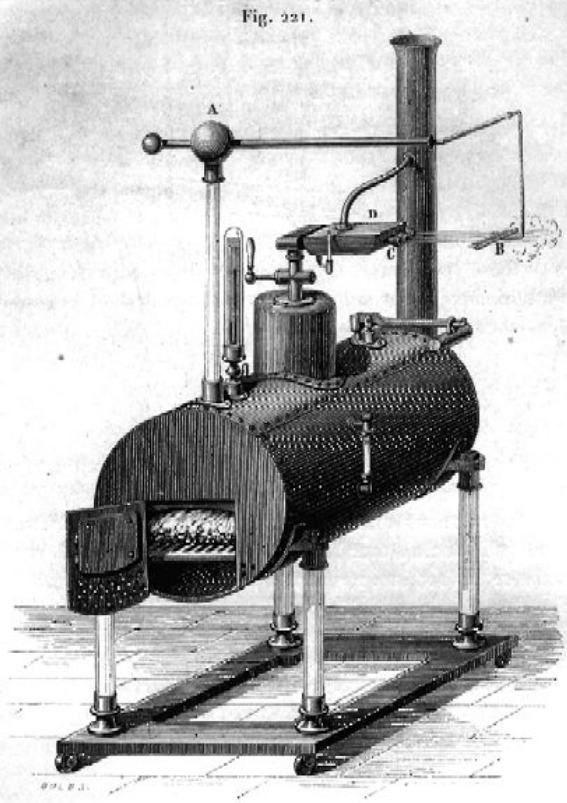


Fig. 222.

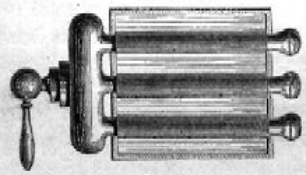
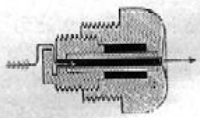


Fig. 223.



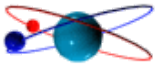
(*fig.* 223) et qu'elle charge négativement en prenant elle-même l'électricité positive. On recueille cette électricité sur un peigne de laiton B qui est isolé. Cette machine atteint une grande énergie. Celle que possède l'Institution Polytechnique de Londres et qui a 2 mètres de longueur, donne 46 jets; elle produit des étincelles de 60 centimètres de longueur, qui éclatent d'une manière à peu près continue.

AIMANTATION PAR L'ÉLECTRICITÉ. — Les procédés dont nous venons de parler sont complètement tombés en désuétude. On emploie aujourd'hui presque exclusivement dans les ateliers les deux méthodes suivantes, basées sur des principes que nous développerons ci-après :

1° On se sert d'un *électro-aimant* fixe (*fig.* 67), sur les pôles duquel on fait glisser, alternativement et en sens contraire, le barreau à aimanter. L'effet de cette friction est de développer à l'extrémité qui abandonne l'aimant un pôle de nom contraire à celui qui agit, d'où l'on voit que les frictions inverses sur les deux pôles tendent à produire une aimantation dans le même sens.

Fig. 68.
AIMANTATION PAR UN COURANT.

2° Quand on veut produire une aimantation assez énergique, l'électro-aimant doit être puissant, et dès lors l'adhérence du barreau est telle, que l'opération devient incommode. D'ailleurs le barreau peut se trouver fortement rayé par la friction. On préfère dans ce cas faire mouvoir le long du barreau une bobine (*fig.* 68) traversée par un courant.

C'est Arago (1) le premier, après les belles découvertes d'Œrsted, qui découvrit que l'on pouvait ainsi faire des aimants puissants, en soumettant l'acier et le fer à l'action d'un courant électrique. Une expérience intéressante a été faite en 1875, dans le même ordre d'idées, à l'Académie des sciences, par M. Thomasi. Ce physicien produit des aimants, non plus en soumettant du fer ou de l'acier, comme Arago, à l'action d'une hélice dans laquelle passe un courant électrique, mais en plaçant un barreau dans les spires d'un tube creux au sein duquel circule un jet de vapeur. Un tube de cuivre de 2 à 3 millimètres de diamètre est enroulé en hélice autour d'un cylindre de fer ; on fait passer de la vapeur à 5 ou 6 atmosphères dans le tube, et le morceau de fer s'aimante et conserve son aimantation pendant toute la durée du passage de la vapeur.

Quelle est la cause du phénomène ? M. Thomasi s'en est tenu aux faits, sans insister sur la théorie. Quelques physiciens ont avancé que la vapeur, en circulant, créait un courant thermo-électrique par suite de la différence de température entre la paroi froide extérieure et la paroi chaude intérieure du tube. Il semble que, s'il en était ainsi, l'égalité de température tendant à s'établir, le courant disparaîtrait vite. Il est plus probable que le frottement des gouttelettes de vapeur d'eau, entraînées contre les parois du tube, crée de l'électricité qui aimante les barreaux.